

Astro Night Planner

Practical guide for planning astrophotography nights with weather, twilight, Moon, object selection, equipment, horizon and framing

Updated: 20 June 2026

Important note on settings, cookies and backup

The Astro Night Planner stores profiles, equipment, locations, horizon profiles and settings locally in IndexedDB. Classic cookies are not the primary storage location.

The installable web app files are stored separately in Cache Storage. Deleting only the browser cache usually does not delete IndexedDB data; deleting site data can remove settings, profiles and stored file permissions.

Persistent storage protects only against automatic browser cleanup, not against deliberate deletion by the user. Regular external backups remain necessary.

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Overview and concept

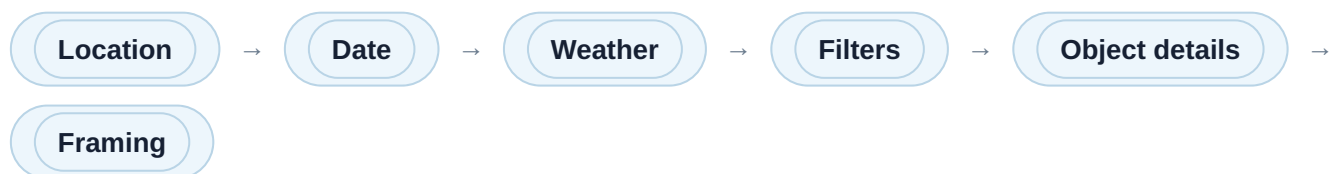
The Astro Night Planner is a practical decision tool for deep-sky astrophotography. It does not only answer whether an object is theoretically visible; it combines visibility, twilight, Moon, weather, personal equipment, location, horizon and framing into one object rating.

Planning

Location, date, temporary profiles, weather, cloud map, object filters, object list, altitude curve, horizon view and Aladin framing.

Settings

Equipment, locations, horizon profiles, display, rating logic, weather profiles, backup, help and legal information.



Benefit: You no longer need to check a weather app, star atlas, framing tool and object list separately. The Planner brings these steps into one workflow.

First steps: recommended setup

- 1 Open **Settings - Equipment** and check telescope, camera and mount.
- 2 Create your imaging location under **Locations & horizon**. Use search or GPS, but verify the coordinates.
- 3 Enter your personal horizon. Even a rough profile is better than assuming a completely flat horizon if trees, buildings or hills interfere.
- 4 Check defaults for weather rating, display and cloud map under **Central settings**.
- 5 Switch to Planning, select date and location, and check weather and cloud map first.
- 6 Filter the object list by catalogs, types, minimum altitude, visible duration, Moon distance and size.
- 7 Open an object and assess altitude curve, horizon view and framing.
- 8 Set up an external backup before creating many profiles and locations.

Example: For a short evening session choose the nautical window, set visible duration to 2 hours, minimum altitude to 25 degrees and Moon distance to 40 degrees. Then inspect the best objects by framing and cloud evolution.

Language and documentation

On first launch the app checks the browser or device language. German starts in German; all other languages start in English. After manual switching, the saved selection is used.

The title bar contains the switch  **DE | EN**. It changes the active language and stores the choice locally. Help and PDF links automatically open the corresponding language version.

Note: The German legal texts are authoritative. English texts are provided as an additional aid.

Local data, cache and backup

Profiles, equipment, locations, horizon profiles and settings are stored in **IndexedDB**. Installable app files are stored separately in **Cache Storage**. Classic cookies are not the primary storage location.

Browser action	Typical effect
Delete only cached images and files	Usually removes PWA files, but not IndexedDB data.
Delete site data	Can remove profiles, settings, file permissions and cache.
Request persistent storage	Protects against automatic cleanup, not deliberate deletion.

Recommendation: Use *Export all data* regularly or enable automatic backup. `astro-night-planner-aktuell.json` is the current state; dated files are historical restore points.

Location, date and planning window

Location and date in Planning apply to the current planning session. The default location is set in Settings. A planning night starts on the evening of the selected date and ends the next morning.

Window	When useful?
Sunset to sunrise	Overview curves and broad object paths.
Nautical planning window	Recommended default for many deep-sky plans because bright twilight edges are reduced.
Astronomical night	For demanding faint targets and very dark conditions.

The displayed curve can be wider than the rating basis. The current label **minimum visibility** technically means a minimum duration above the minimum altitude and personal horizon. It will later be renamed more clearly.

Profiles for this plan

This section allows temporary deviations without overwriting saved defaults. This is useful when testing another telescope, camera, horizon profile or weather profile for one night.

- **Telescope/camera:** affect field of view, framing and framing rating.
- **Horizon profile:** affects visible duration and horizon view.
- **Imaging quality profile:** affects wind and gust evaluation.
- **Display profile:** controls object-list columns, order and page size.

Practice: On a windy night you can temporarily test a more robust wind profile without changing the normal default.

Weather and imaging quality

The app combines weather models into a consensus. Large KPI fields summarize the night; the hourly table shows when conditions change.

Criterion	Meaning for imaging
Clouds	Main exclusion factor. High clouds can strongly degrade transparency.
Effective transparency	Combines cloud cover and atmospheric clarity.
Seeing	Tendency from jet stream and surface wind; not an exact arcsecond measurement.
Wind/gusts	Expected imaging quality, not mechanical safety.
Dew gap	A small gap increases dew risk.
Moon	Influence via distance, illumination and altitude.

Interpretation: A good total score does not replace checking individual values. A 75/100 night may be good for narrowband but poor for faint galaxies due to Moon or transparency.

Animated cloud map

The cloud map shows cloud distribution, model disagreement and optional precipitation layers. **Map + clouds** uses a dark topographic map for orientation. **Clouds only** shows the same data on a neutral dark background and is best for judging cloud structure.

- Percent values are shown at the actual forecast points.
- Smoothing changes only the display, not the underlying weather data.
- 15, 30 or 60 minute steps interpolate between hourly model values.
- Precipitation, rain and snow can be displayed as additional context.

Example: If the value at your location is good but the map shows a cloud edge just west of you, uncertainty may be higher than the single number suggests.

Meteoblue as a reference source

Meteoblue is used as an independent reference view and is not automatically included in the model consensus. It is useful when the Open-Meteo models disagree or when you want an additional check of seeing and upper-wind tendencies.

Use Meteoblue especially before long trips or on nights when the cloud map shows a sharp cloud boundary.

Combining object filters

Filters should reduce the object list quickly to useful targets. Start broad and tighten only afterwards.

Filter	Recommendation
Catalogs	Activate only catalogs you really want to see. Special catalogs can create very large result sets.
Object types	Narrowband: emission nebulae, planetary nebulae, supernova remnants. Broadband: galaxies, clusters, reflection nebulae.
Minimum altitude	25 degrees is a good starting point. Lower can work with a free horizon; use more with light domes or obstacles.
Minimum visibility	Prevents targets that are usable only briefly. It means minimum duration above horizon/minimum altitude.
Moon distance	Be strict for faint broadband targets, more relaxed for narrowband.
Object size	Helps find targets that match your field of view.

Narrowband practice: Catalogs NGC/IC/Sh2, types emission nebulae and supernova remnants, filters Ha/OIII/SII, moderate Moon distance, then check framing.

Object list, score and mini altitude profiles

The object list is sorted by score. The score combines weather, altitude, Moon, visible duration and other criteria according to the active profile. Mini altitude profiles show immediately whether an object is useful for a longer period or only briefly near the edge.

Columns can be shown, hidden and sorted per display profile in Settings. Compact profiles are useful on mobile; the detailed view can show more information on desktop.

Do not blindly trust rank 1: Always open details for your favorites. Framing, object size and cloud evolution can change the practical decision.

Object details, altitude curve and horizon

Clicking an object row opens details below. Altitude curve and horizon view use the same time control, so altitude, azimuth, twilight, horizon clearance and weather values can be assessed together.

The horizon view is essential if your local horizon is not flat. An object may be astronomically 30 degrees high but still behind a tree or building.

Example: A target with a high score shortly after twilight may be unusable if it clears your local obstruction only above 35 degrees in that direction.

Framing with Aladin Lite

The Aladin view shows the sky image, camera frame and - where available - a curated object outline. If no outline is available, the app falls back to the technical ellipse based on catalogue size and position angle. Camera frame, comparison frames, object outline, Moon marker and labels are overlay layers above the native Aladin survey and follow zooming and panning.

- **Frame to image center:** places the camera frame on the currently visible center after panning.
- **Reset object orientation:** restores the technical ellipse to the catalogue position angle if an object has no curated outline.
- **Time in planning window:** does not change the star field; it updates time-dependent values such as altitude, direction, twilight, weather and the optional Moon position.
- **Show object names:** displays labels depending on zoom level.
- **Open sky image in new tab:** opens the Aladin detail map larger in a separate browser tab.

Why the star field does not rotate: In an equatorial framing context, object coordinates and camera angle remain fixed relative to the sky. Time mainly changes altitude, azimuth and observing conditions.

Aladin surveys, setups and measuring tool

Use the native survey rendering

Aladin surveys are loaded in their native rendering. A survey such as **DSS2 red** must therefore not be artificially colored red: the name describes the red photographic band, while Aladin's native visual presentation is black and white. The Planner does not apply an artificial colormap to monochrome surveys. Surveys appear in color only when the source itself provides a color survey or color composite.

Example survey choice: Galaxies usually work well with an optical color survey or DSS2 red. Emission nebulae can be easier to judge in H-alpha. Dust and dark nebulae may benefit from infrared or dust surveys. When adding a survey, always check whether the wavelength range really answers your planning question.

Add custom surveys

Under **Settings - Central settings - Display** you can choose which Aladin surveys appear in the object map. You can also add custom HiPS surveys manually. A display name and the technical HiPS ID or a suitable HiPS URL are required. Custom surveys remain stored locally in your profile.

Field	Recommendation
Display name	Use a short, clear name, for example <i>NSNS H-alpha</i> or <i>My Ha survey</i> .
HiPS ID	Use the exact survey ID or URL. Invalid IDs prevent Aladin from loading an image.
In selection	Only enabled surveys appear in the compact object-map selection.

Setup combinations and comparison frames

Planning uses one active setup consisting of telescope and camera. This setup controls field of view, pixel scale, framing rating and filters such as *fits on sensor*. In the Aladin detail view you can additionally show up to three setup combinations as local comparison frames. These frames do not change the global object list and only affect the currently opened sky image.

Configuration example: Create *Askar 200 + QHY268M* as your standard setup. Add a second setup, for example *Lacerta + QHY268M*. Planning still uses one active setup; in Aladin you enable the second setup only as a comparison frame to see immediately whether the object fits better at another focal length.

Measuring tool in the external tab

The external Aladin tab includes a measuring tool. After two clicks it shows the angular distance numerically and as a visible line with endpoints in the sky image. This is useful for object separations, rough mosaic planning, Moon distance checks and comparing neighbouring nebula structures.

Practice tip: Use the embedded map for quick framing and the external tab when you want more space for measuring, checking surveys or inspecting details.

Equipment

Telescopes, cameras and mounts are stored locally. Telescope and camera determine field of view, image scale and framing rating. The mount is currently documented and may later be used for more detailed quality profiles.

Field	Why important?
Focal length	Determines field of view together with sensor size.
Sensor width/height	Determines whether an object fits into the camera frame.
Pixel size	Basis for sampling and future detail ratings.

Locations and horizon profiles

Locations contain coordinates, elevation, time zone and horizon profiles. Search results should be checked deliberately, especially for postal codes or places with identical names. The time zone is stored as an IANA time zone.

Several horizon profiles per location can be useful: garden, terrace, balcony, mobile field or observatory. A profile can be selected temporarily in Planning without changing the default.

Practice: A location can be free to the south but blocked by trees to the north-east. This strongly affects M31, M45 and early autumn targets.

Central settings and defaults

Central settings define defaults, not necessarily the current plan. Many values can be temporarily overridden in Planning.

- **Quality traffic light:** default red 0-59, yellow 60-79, green 80-100.
- **Deep-sky weighting:** default: clouds 30%, transparency 15%, seeing 10%, wind 10%, dew 10%, Moon 10%, altitude 10%, duration 5%.
- **Wind profiles:** describe expected imaging quality for different setups.
- **Cloud map:** default mode, smoothing, grid and percent values.
- **Display profiles:** object-list columns, order and page size.

Saving: Many settings sections apply changes only after saving. This protects against accidental permanent changes.

New filters, target list and settings tabs

The object filters have been extended so the list can better match the actual imaging situation. Many values are split into persistent defaults in Settings and temporary choices in Planning. This lets you adapt a single night without changing your normal setup.

Minimum score

The **Minimum score** field hides objects below a given quality value. The default is **0**, meaning no restriction. A value of 60 keeps only targets that reach at least the yellow quality range.

Minimum visible duration

This is the minimum time during which an object is above the minimum altitude and your personal horizon. Its reference period can be selected separately. The default is the nautical planning window so bright edge periods are not accidentally counted.

Object size in degrees and arcminutes

Minimum and maximum object size are entered with separate degree and arcminute fields. The app still calculates internally in arcminutes. Minute values above 59 are prevented or normalised. Size presets can be stored in Settings and adjusted temporarily in Planning.

Example: The default range 0°03' to 4°00' removes very tiny specialist objects while keeping many classic deep-sky targets for medium focal lengths.

Surface brightness and missing values

The surface-brightness filter helps identify very faint extended targets. Because not every catalogue object has reliable data, objects without a surface-brightness value remain visible by default. Strict filtering can hide them deliberately.

Imaging filters and object-type profiles

The recommended filters L, R, G, B, Ha, OIII and SII can be used as a multi-select filter. The additional option **no filter information** prevents incomplete catalogue entries from disappearing unnoticed. Object-type combinations can be saved as profiles, for example galaxies, nebulae or narrowband targets. Manual changes in Planning remain temporary.

Framing filter

The framing filter distinguishes well-framed, near-edge, too-large and unknown objects. It uses the active telescope/camera setup and is helpful when you want targets that fit without a mosaic.

New catalogues and My imaging targets

Barnard and LDN/LBN can be enabled as additional catalogues; they are initially inactive. The personal catalogue **My imaging targets** is also initially inactive. You can mark objects, add notes, priority, status and reference links. When this is the only active catalogue, Planning answers the practical question: *What can I image tonight that I have always wanted to capture?*

Moon in the object map

The Moon can optionally be displayed in the Aladin view. Its position is calculated for the selected time in the planning window and updates when the time slider moves. It is initially disabled to keep the framing view clean.

Settings tabs

Settings are split into tabs. Central settings, locations/horizon, info/help/backup and My imaging targets are easier to navigate. The object-list configurations Compact, Standard and Detailed remain visible and no longer collapse automatically when columns are moved.

Export, import and restore

Export all data includes all profiles, equipment, locations and settings. **Export current profile** transfers a single profile. **Import profile** creates a new imported profile and does not overwrite existing profiles without confirmation.

- 1 Create a full backup regularly.
- 2 Store the file outside the browser profile.
- 3 Use `astro-night-planner-aktuell.json` for normal recovery.
- 4 Use dated files only when deliberately returning to an older state.
- 5 After restore, check active profile, location and equipment.

PWA installation and updates

The app can be installed as a PWA. App files are available locally; weather, maps, catalog updates and external sky images still require an internet connection.

When a new version is available, an update banner may appear. After updating, fully closing and reopening the installed PWA can be necessary. Deleting site data is not needed for normal updates.

Legal notice, privacy and usage notes

The footer provides access to legal notice, privacy, usage notes, data sources & licenses, help and version. The app does not integrate its own analytics, tracking or error reporting. GitHub Pages and external services can process technically required connection data.

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Weather and calculation values are planning aids. Safety decisions must be based on local conditions and official warnings.

Troubleshooting

Problem	Recommendation
Old display or old help	Reload the app, accept the update prompt, fully close and reopen the installed PWA.
Weather data missing	Check internet connection and try later; external services can be temporarily unavailable.
Map does not appear	Repeat date change, collapse/open the map section or reload the page.
Aladin does not load	Reload sky image, check network, try the external service later.
Profiles are missing	Check whether site data was deleted; restore the latest external backup.

Glossary

Term	Meaning
IndexedDB	Browser database for profiles and settings.
Cache Storage	Storage for PWA app files.
Nautical window	Period with the Sun below about -6 degrees.
Seeing	Atmospheric turbulence that broadens or distorts stars.
Effective transparency	Transparency rating including cloud cover.
Framing	Fit of object size and camera field.
Minimum visibility	Current label for minimum duration above minimum altitude and horizon; it will be renamed later.

Astro Night Planner 1.0 - User manual. This manual describes how to use the app and does not replace checking actual local conditions.